

## Defining the variable

Model: package masses are approximately normal with mean 500 g and standard deviation 20 g.

**What does  $X$  represent if we select one package?**

## A lower-tail probability

Model:  $X \sim N(500, 20)$ , measured in grams.

**What is the probability that one package weighs less than 480 g?**

## An upper-tail probability

Model:  $X \sim N(500, 20)$ , measured in grams.

**What is the probability that one package weighs more than 530 g?**

## A probability between cutoffs

Model:  $X \sim N(500, 20)$ , measured in grams.

**What is  $P(490 < X < 520)$ ?**

## Outside an acceptable range

Model: package masses follow  $N(500, 20)$ . The accepted range is 470 to 530 g.

**What proportion falls outside the accepted range?**

## Strict or inclusive?

Model: exact package mass has a continuous normal distribution.

**Does  $P(X < 480)$  differ from  $P(X \leq 480)$ ?**

## A technology input

A normal cumulative probability tool needs a mean, standard deviation, and cutoff or interval.

**For  $P(X > 530)$  when  $X \sim N(500, 20)$ , what information should you enter?**

## Checking the direction

Model:  $X \sim N(500, 20)$ . A calculation for  $P(X > 530)$  returns 0.9332.

**Is that result plausible for a cutoff above the mean?**

## **From probability to expected count**

Model: 8% of packages fall outside specification. A shipment contains 250 packages.

**How many out-of-specification packages should we expect?**

## A percentile

A percentile cutoff specifies area at or below that value.

**What probability equation defines the 90th percentile  $k$ ?**

## Finding a percentile

Model:  $X \sim N(500, 20)$ , measured in grams.

**What mass is the 90th percentile?**

## An upper-tail cutoff

Model:  $X \sim N(500, 20)$ , measured in grams.

**Only 5% of packages exceed what mass?**

## A lower-tail cutoff

Model:  $X \sim N(500, 20)$ , measured in grams.

**What mass separates the lowest 10% from the rest?**

## The middle 90%

Model:  $X \sim N(500, 20)$ , measured in grams.

**What symmetric interval around the mean contains the middle 90%?**

## Quartiles from a model

Model:  $X \sim N(500, 20)$ , measured in grams.

**What masses are the first and third quartiles?**

## An interquartile range

Model:  $X \sim N(500, 20)$ , measured in grams.

**What is the model's interquartile range?**

## Different spreads

Model A is  $N(500,10)$ . Model B is  $N(500,20)$ , with mass in grams.

**Which model gives a greater probability of mass above 530 g?**

## Different centers

Model A is  $N(500, 20)$ . Model B is  $N(510, 20)$ , with mass in grams.

**Which model gives a greater probability of mass below 480 g?**

## **A probability or a cutoff?**

A manager asks for the mass exceeded by only 2% of packages.

**Does the requested answer have units of grams or units of probability?**

## Conditional normal probability

Model:  $X \sim N(500, 20)$ . A package is known to weigh more than 500 g.

**What is the probability that it weighs more than 520 g?**

## A model boundary

A normal model assigns some probability to every real-valued interval.

**When could its prediction of negative durations make it unsuitable for duration data?**

## One package or an average?

Model: independent package masses follow  $N(500, 20)$ .

**Should an average of 100 masses be analyzed using the same standard deviation as one mass?**